

SWYNEX – Task 2

Exploratory Data Analysis Report

Dataset: Cleaned HR Employee Dataset from Task 1

1. Objective

Explore the cleaned dataset, calculate statistics, identify trends and patterns, create charts, and communicate useful insights.

2. Dataset Overview

The analysis uses 500 cleaned employee records. Findings are specific to this controlled demonstration sample.

3. Important Statistics

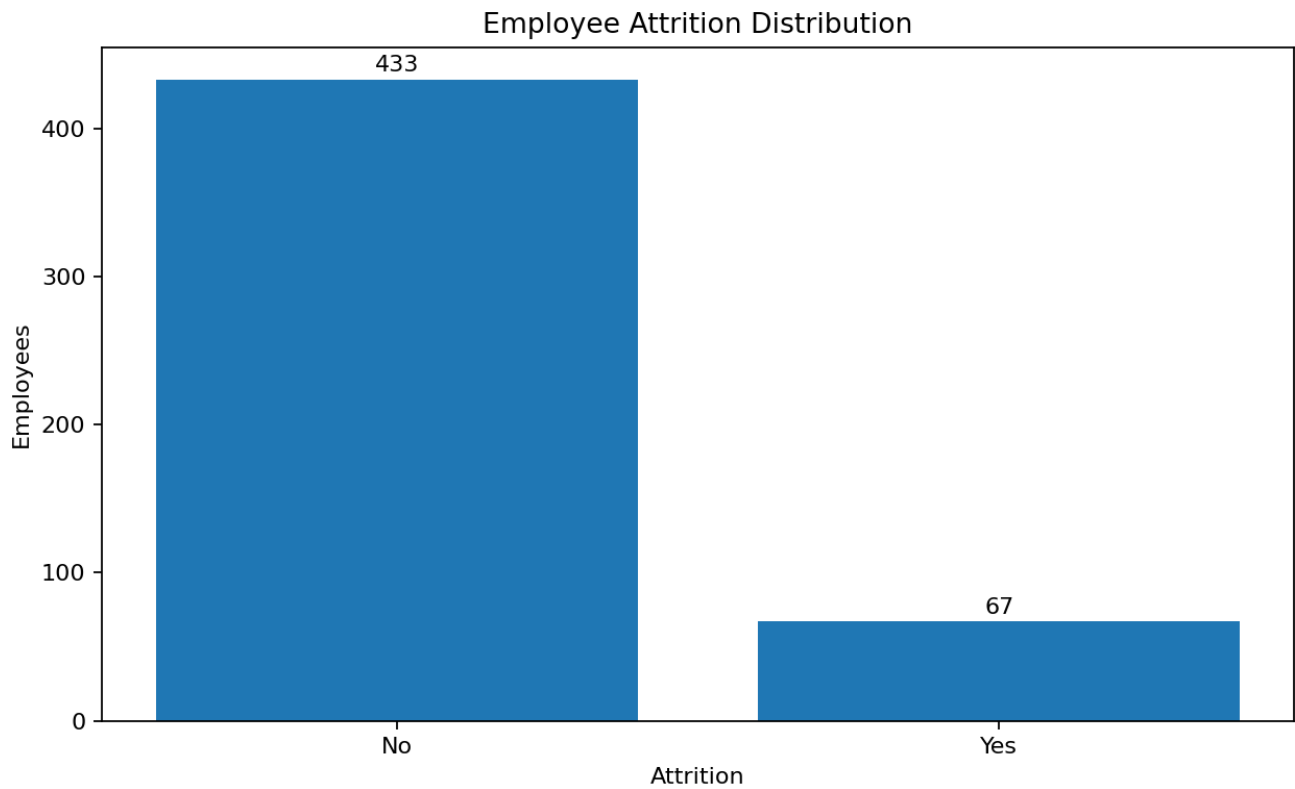
Metric	Value
Total employees	500.0
Attrition cases	67.0
Employees retained	433.0
Overall attrition rate (%)	13.4
Average age	36.46
Average monthly income	6643.35
Median monthly income	6514.0
Average years at company	6.28
Average job satisfaction	2.57
Average work-life balance	2.5

4. Key Insights

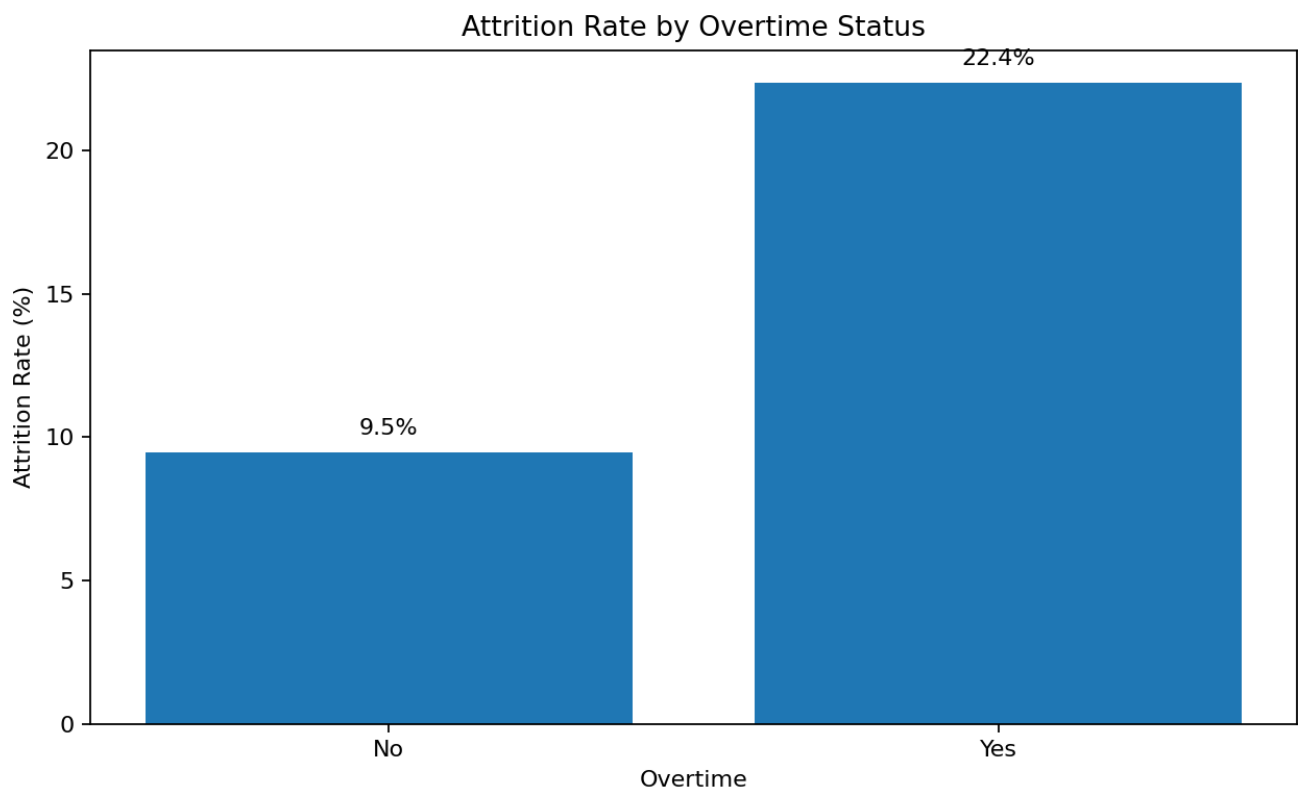
1. The cleaned sample contains 500 employees, with 67 attrition cases. The overall attrition rate is 13.4%.
2. Overtime is the clearest segmentation in this sample: employees working overtime have a 22.4% attrition rate versus 9.5% for employees without overtime.
3. Human Resources has the highest department-level attrition rate at 25.0%, compared with 17.4% across the three departments.
4. The Human Resources role has the highest observed role-level attrition rate (37.5%), but it represents only 8 employees, so the rate should be interpreted cautiously.
5. The 26-35 age group has the highest attrition rate at 16.3% in this sample.
6. Employees who left had a lower average tenure (5.33 years) than employees who stayed (6.42 years), a difference of 1.09 years.
7. Median monthly income is very similar for employees who stayed (6,514) and those who left (6,469), suggesting income alone is not a strong separator in this sample.
8. Work-life balance level 4 shows an 8.8% attrition rate versus 16.1% at level 2, indicating a possible relationship worth validating with a larger dataset.

5. Visualizations

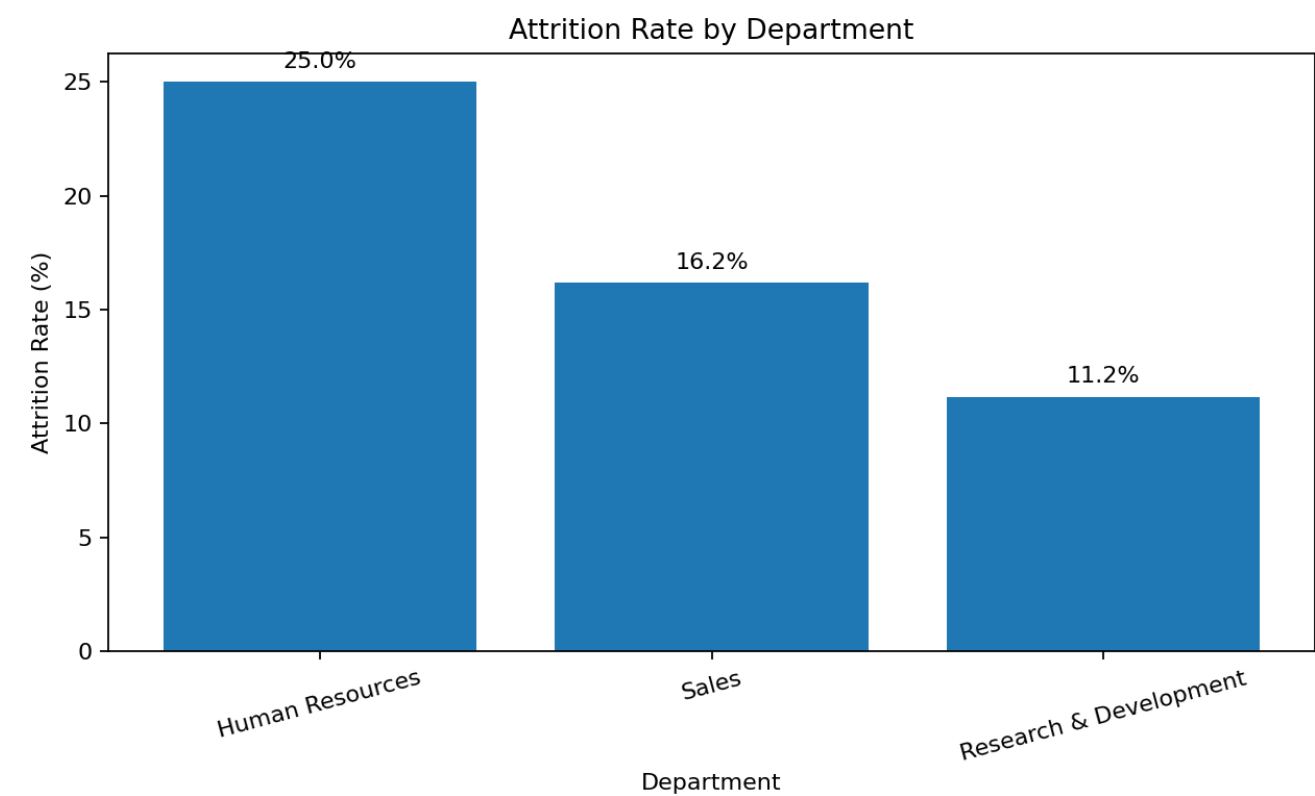
01 Attrition Distribution



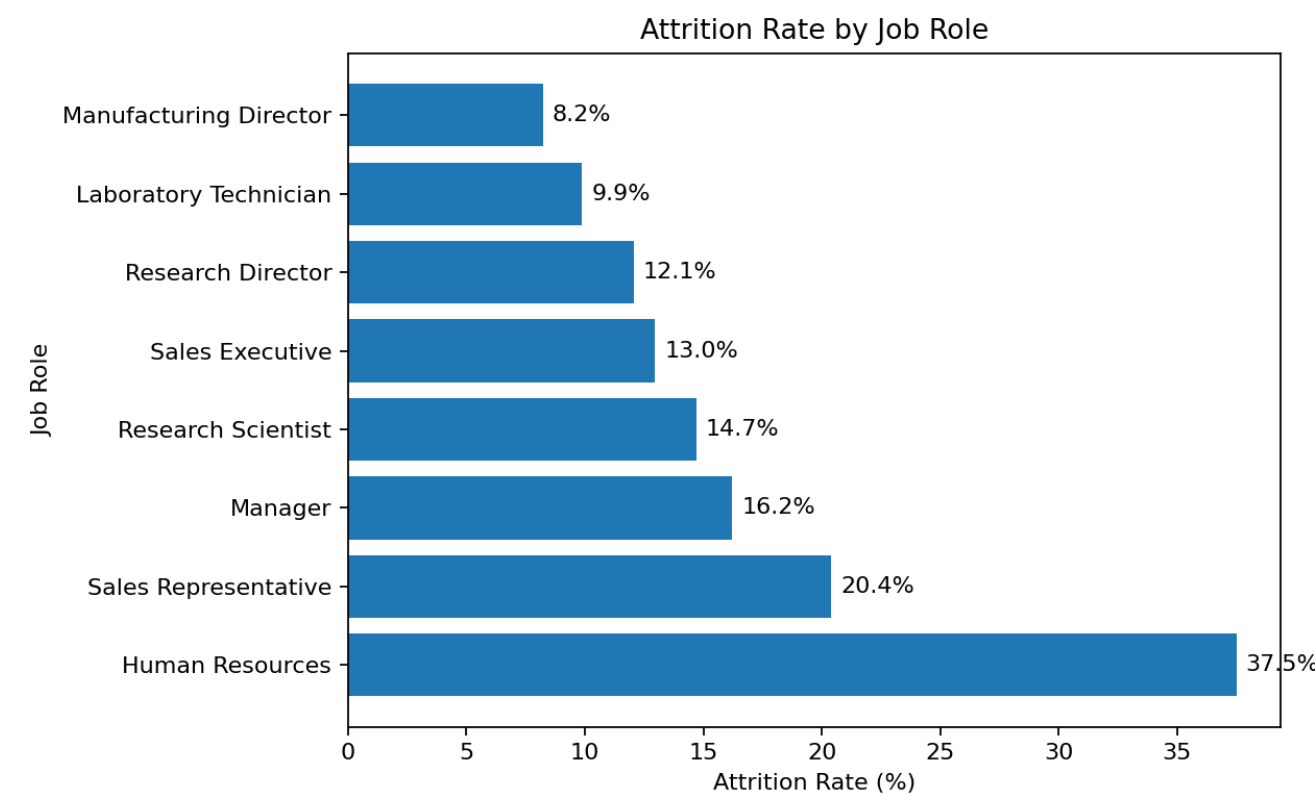
02 Overtime Attrition



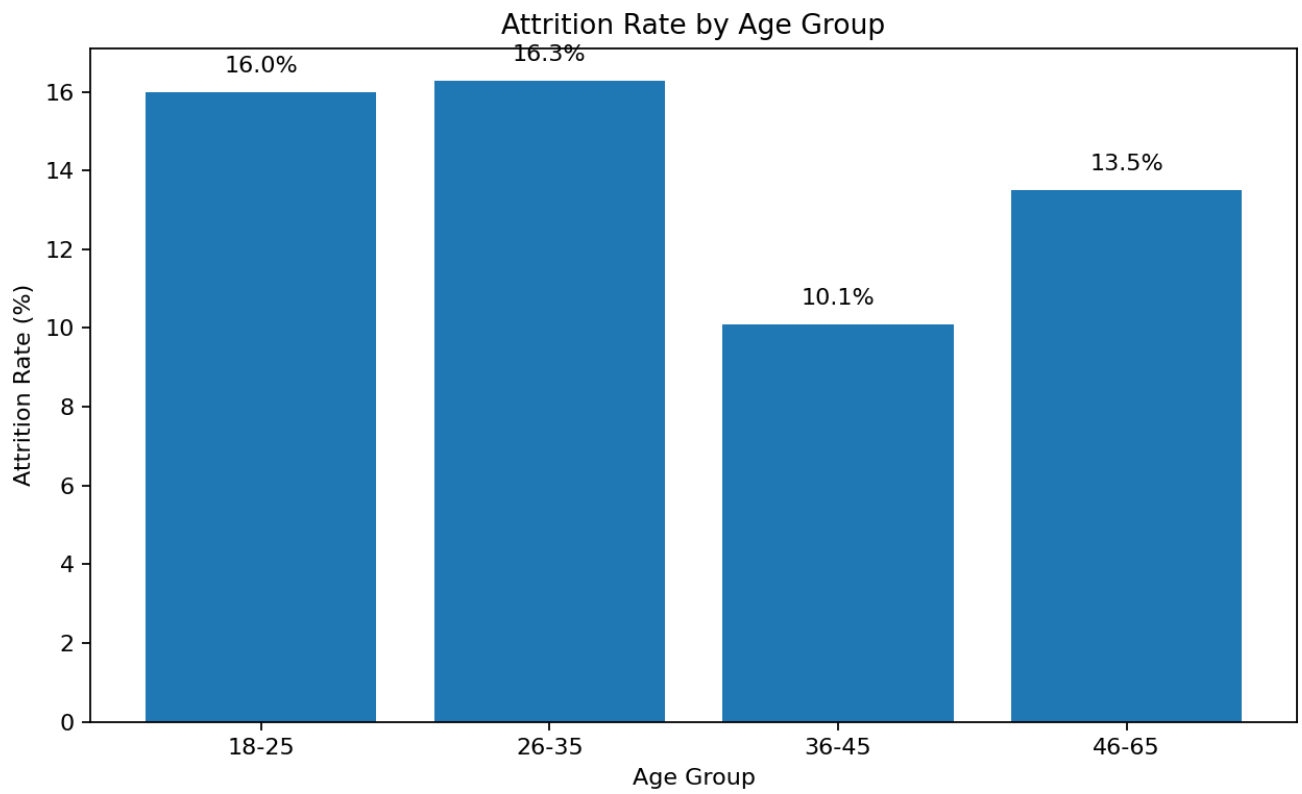
03 Department Attrition



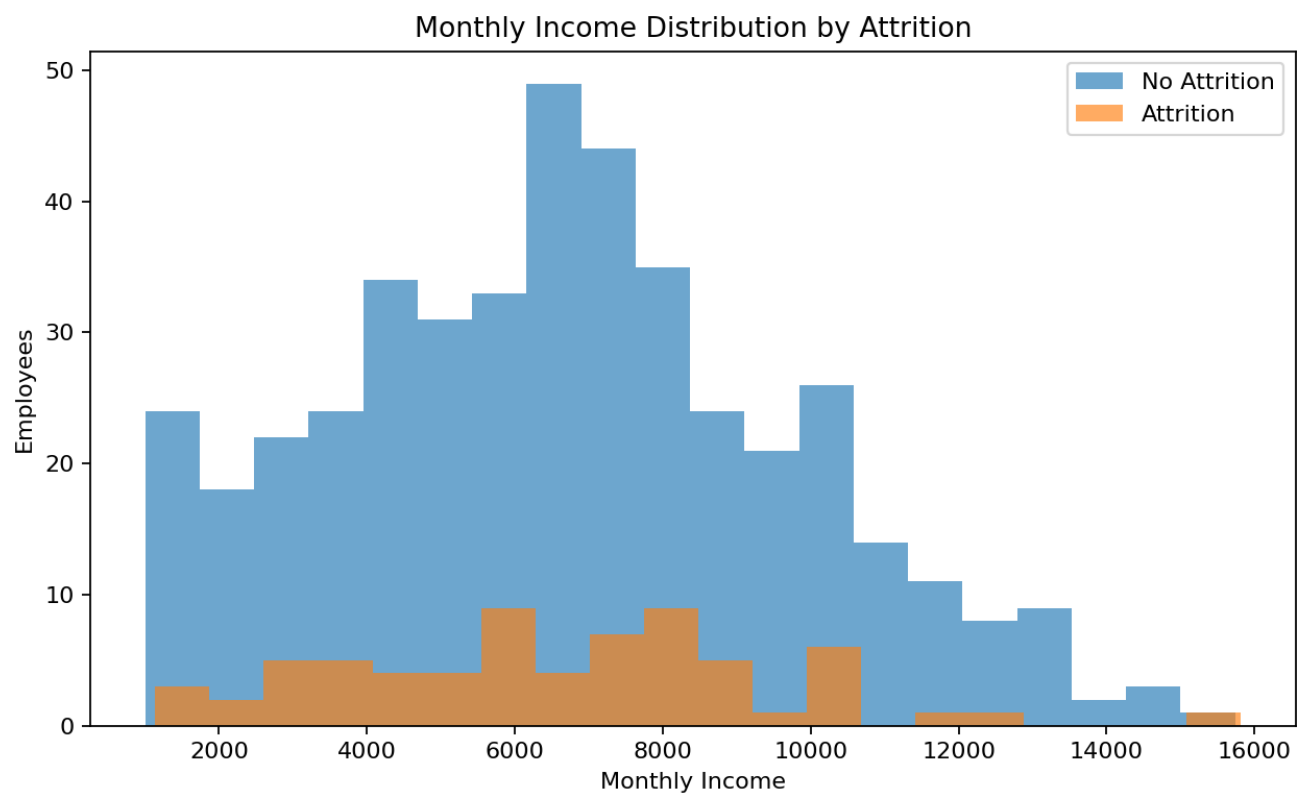
04 Jobrole Attrition



05 Agegroup Attrition



06 Income Distribution



07 Tenure Attrition

Average Years at Company by Attrition

